```
SetFactory("OpenCASCADE");
Merge "Gemini 5.step";
//+
Physical Surface(78) = {20, 16, 21, 17, 19, 25, 23, 24, 22, 18, 15, 28, 14, 6, 13, 11, 10, 12, 3, 27, 7, 5, 4, 9, 8, 2, 26, 1};
//+
Physical Volume(79) = {1};



// 1. HEAL & MERGE VOLUMES
// This is critical to turn the 'hollow' shell into a solid piece
all_vols[] = Volume { : };
BooleanFragments{ Volume{all_vols[]}; }{ }

// 2. SET COARSE MESH SIZE
// Increase these numbers to make the mesh faster.
// Try Min=5 and Max=10. If it's still too slow, make them bigger.
Mesh.MeshSizeMin = 1;
Mesh.MeshSizeMax = 2;

// 3. STABLE SURFACE SELECTION
all_surfs[] = Surface { : };

// 4. THE TRANSFINITE LOOP
For i In {0 : #all_surfs[]-1}
   // We attempt Transfinite for the grid look
   Transfinite Surface {all_surfs[i]};
   Recombine Surface {all_surfs[i]};
EndFor

// 5. VOLUME FILLING (Fixes the "Hollow" problem)
```

```
// This tells Gmsh to fill the 3D space.
// Using Algorithm 1 (Delaunay) for the fastest 3D results.
Mesh.Algorithm = 6;     // 2D: Frontal-Delaunay for Quads
Mesh.Algorithm3D = 1;   // 3D: Delaunay (fastest for volumes)
Mesh.RecombineAll = 1;  // Allow some triangles to speed up the process

// Define Physical Group so the solid is recognized
Physical Volume("Solid_Impeller", 80) = {all_vols[]};
```